EDUCATION AS A POSITIVE EXTERNALITY: GOVERNMENT INTERVENTION AND OUTCOMES

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ABSTRACT

This paper explores the role of education as a positive externality and evaluates government intervention outcomes within Nigeria's macroeconomic framework. Using quantitative methods specifically Exploratory Data Analysis of secondary time-series data. It reviews the relationship between public education expenditure, primary school enrollment, and economic growth from 1987 to 2024. The study contrasts the theoretical Pigouvian justification for state subsidies with the practical realities of Nigeria's chronic underfunding and systemic supply-side barriers. Findings reveal that while absolute nominal spending on education has increased, the expenditure as a percentage of GDP remains chronically low, failing to internalize positive externalities or correct the nation's severe human capital deficit. The paper recommends overhauling spending mechanisms and aligning budgetary allocations with international benchmarks to improve socio-economic spillovers and long-term economic growth in Nigeria.

Keywords: ***Positive externality, Government expenditure, Human capital development, Pigouvian theory, Macroeconomic growth***

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1.1 Introduction

The pursuit of education is often viewed through the lens of individual advancement when private investment in human capital is designed to yield higher lifetime earnings. However, from an economic perspective, the classroom extends far beyond the student. Education is a classic example of a positive externality, where the social benefits of consumption significantly exceed the private benefits accrued by the individual.
When an individual becomes educated, the spillover effects ranging from increased innovation and civic engagement to reduced crime rates which benefits society at large, often without the individual being compensated for providing these advantages. This chapter introduces the theoretical framework of education as an externality and outlines why market forces alone fail to reach an optimal level of schooling.

1.2 Statement of the problem

Despite the theoretical consensus that education produces significant positive externalities, Nigeria presents a compelling case study of persistent market failure and sub-optimal human capital accumulation. In a perfectly competitive framework, individuals under-consume education because they base their investment decisions strictly on their Marginal Private Benefit (MPB) such as expected future wages ignoring the Marginal Social Benefit (MSB) that accrues to the wider society, including civic stability, health improvements, and aggregate productivity gains (Adunola, 2011).

To correct this inherent market failure and shift the economy toward a Pareto efficient allocation, the Nigerian government has intervened through initiatives like the Universal Basic Education (UBE) Act of 2004. The UBE was designed to internalize these externalities by mandating free and compulsory basic schooling, theoretically lowering the private cost of education to align with societal needs. However, a stark divergence remains between policy formulation and actual macroeconomic outcomes. High opportunity costs (such as the reliance on child labor in impoverished households) and severe supply-side barriers including chronic underfunding, decaying infrastructure, insecurity in certain regions, and teacher shortages ensure that the private burden of schooling continues to outweigh its perceived returns for many Nigerian families (Aluede, 2006).

Consequently, Nigeria suffers from a profound human capital deficit, characterized by an out-of-school population estimated between 10 to 20 million children, ranking among the highest globally. The World Bank's Human Capital Index (HCI) currently places Nigeria at 0.36, indicating that a child born in Nigeria today will only be 36% as productive in adulthood as they could be if they enjoyed complete education and full health. This gap between the current human capital accumulation and the socially optimal level represents a staggering deadweight loss to the nation's economy (UNICEF, 2024).

Furthermore, analyzing this deficit through an overlapping generations (OLG) framework reveals its dynamically destructive nature. When the current generation under-invests in education due to systemic policy failures and socioeconomic barriers, it establishes a low technological baseline, diminishes wealth transfers, and creates a productivity trap for the subsequent cohort. This failure to capture and internalize educational externalities not only fuels contemporary issues like the "brain drain" of qualified professionals but fundamentally restricts Nigeria's long-term trajectory toward a high-growth, diversified economy.

1.3 Research Objectives:

To assess the impact of government expenditure on educational

To analyze the relationship between education and broader social outcomes

To analyze historical trend of government government expenditures on education

2. Literature Review and Theoretical Framework

2.1 Theoretical Literature review

Economic literature provides several frameworks that explain market failures, the necessity of government intervention, and the long-term growth trajectories of nations. Three pivotal theories in this domain are the Theory of Externalities (Pigouvian Theory), Human Capital Theory (Schultz & Becker), and the Endogenous Growth Theory (Romer & Lucas).

1. The Theory of Externalities (Pigouvian Theory)

Developed by Arthur C. Pigou in the early 20th century, the Theory of Externalities provides the classic economic justification for government intervention in the free market. An externality occurs when the production or consumption of a good imposes unintended costs (negative externality) or confers unintended benefits (positive externality) on third parties who are not directly involved in the transaction. Ihugba, et al. (2019) examined the impact of government education expenditure on primary school enrollment in Nigeria using the Autoregressive Distributed Lag (ARDL) bounds testing technique. The study established a long-run equilibrium relationship between public education expenditure and primary school enrollment, with an Error Correction Model (ECM) indicating a substantial 88% speed of adjustment to equilibrium within one year. Despite this long-run linkage, short-run dynamics showed that government education expenditure had an insignificant effect on actual primary school enrollment, suggesting that Pigouvian subsidies aimed at reducing under-enrollment may be diluted or ineffective in the short-term context of Nigeria’s education sector.

2. Human Capital Theory (Schultz & Becker)

Pioneered by Theodore Schultz (1961) and extensively developed by Gary Becker (1964), Human Capital Theory revolutionized how economists view labour. Schultz and Becker argued that individuals invest in themselves through education, training, and healthcare in the same way a business invests in physical machinery. According to this theory, education is not mere consumption; it is an investment that increases an individual's cognitive and productive capacities. This accumulation of human capital directly enhances the marginal productivity of the worker, which in turn commands higher wages in the labor market. At the macroeconomic level, the aggregate accumulation of human capital within a population is posited as a primary driver of national economic development and poverty reduction. A study by Oladipupo et al. (2023) titled *"Determinants of Human Capital Development in Nigeria"* relied heavily on this theoretical framework. The authors examined the impact of government expenditure on education and health on the Human Development Index (HDI) from 1981 to 2023. Grounded in Becker’s assertions, the study utilized an Error Correction Model to empirically prove that increases in educational spending and healthcare investment have positive, statistically significant effects on workforce productivity and human development in Nigeria, thereby validating the theory that continuous investment in human capital is an economic prerequisite.

3. Endogenous Growth Theory (Romer & Lucas)

While traditional neoclassical growth models (like the Solow-Swan model) attributed long-term economic growth to exogenous (external) technological progress, the Endogenous Growth Theory, developed by Paul Romer (1986) and Robert Lucas (1988), argues that economic growth is generated from *within* a system. The theory places human capital, innovation, and knowledge at the absolute center of macroeconomic growth. A critical departure from previous models is the concept of "increasing returns to scale." Romer and Lucas argued that unlike physical capital, which is subject to diminishing returns, knowledge and ideas have positive spillover effects (externalities). When an economy invests in education, the resulting knowledge can be shared infinitely without being depleted, making surrounding workers and industries more productive. Therefore, government policies that permanently increase investments in education and research can lead to a permanent increase in the economic growth rate. A recent empirical study by Danjuma et al. (2025) the study found that expenditure on education may not show immediate short-term impacts, it demonstrates substantial and sustainable positive effects on Nigeria's economic growth in the long run. The authors attributed this to the gradual improvement in workforce quality and innovation capabilities, directly echoing Romer and Lucas's theoretical predictions.

2.2 Empirical Literature

The empirical literature surrounding education as a positive externality demonstrates that while human capital accumulation theoretically drives socio-economic development, the actual outcomes of government intervention vary significantly based on policy implementation, institutional quality, and geographic context.

Studies such as Chima and Yusuf (2024) utilized an Autoregressive Distributed Lag (ARDL) methodology to analyze the relationship between educational funding and economic performance. The study incorporated variables such as government expenditure on education, labor productivity, and economic growth, with a scope covering Nigeria from 1990 to 2023. Their findings revealed that public expenditure on education actually had a negative and statistically insignificant impact on both short- and long-run growth due to systemic inefficiencies and misallocations. Consequently, they reached the conclusion that a total reevaluation of government spending mechanisms is required to capture the true externalities of human capital investment.

Similarly, Aribaba and Ahmodu (2022) employed Ordinary Least Squares (OLS) and Spearman’s rho correlation methodology to examine the effectiveness of educational budgets. Analyzing variables including government recurrent expenditure, capital expenditure, human capital development, and economic growth, the scope of their research spanned Nigeria's democratic era from 1999 to 2021. The findings indicated that while recurrent expenditure had positive effects on growth, the overall capital educational budget had a low and statistically insignificant impact. The authors formed the conclusion that those responsible for educational budgets must strictly strengthen implementation monitoring to ensure funds translate into tangible economic externalities.

Beyond aggregate GDP, the positive externalities of education are deeply felt at the household level. Okwu et al. (2022) adopted an ARDL and Granger Causality methodology to evaluate

the social impacts of state funding. Their study utilized variables such as the education budget, health budget, inflation, and poverty reduction (proxied by life expectancy), with a scope covering Nigeria from 1980 to 2019. The findings demonstrated that government education spending correlates positively and significantly with poverty reduction in the long run. They reached the conclusion that improved state access to quality education remains the most effective mechanism for breaking intergenerational cycles of poverty.

Addressing the expectations of graduates, Nwadike et al. (2023) utilized a quantitative survey and T-test methodology to assess advanced degrees. They measured variables such as perceived social returns, actual monetary returns, and university type, within a geographic scope limited to Anambra State, Nigeria. The findings highlighted a massive, statistically significant gap between the expected returns of Master's/Ph.D. graduates and their actual socio-economic outcomes, largely driven by Nigeria's high unemployment rate. The authors formed the conclusion that harsh macroeconomic realities frequently distort and suppress the expected microeconomic returns to education.

Despite proven benefits, systemic barriers hinder optimal outcomes. Odigwe and Owan (2019) employed trend analysis and ANOVA methodologies to evaluate state commitment to education. They tracked variables comprising national budgetary allocations, GDP, and the UNESCO 26% funding benchmark, with a scope comparing Nigeria to other West African nations over a multi-decade trend. The findings exposed that Nigeria consistently allocates between 5-8% of its budget to education, significantly below the UNESCO benchmark and regional peers. The conclusion emphasized that this chronic inadequate funding neutralizes the intended positive externalities of educational policies like the Universal Basic Education (UBE) program.

Looking at infrastructural prerequisites, Salisu (2023) applied an ARDL methodology to study educational outputs. The research measured variables including public education funding, infrastructural development indices, and economic performance, with a scope encompassing modern Nigeria. The findings indicated that education acts as a catalyst for technological adaptation only when matched with adequate physical infrastructure. The conclusion was that the government must view capital infrastructure and educational spending as complementary interventions to capture positive spillovers.

2.3 Theoretical framework

The Pigouvian Theory of Welfare Economics (Theory of Externalities)

The Pigouvian Theory of Welfare Economics provides a fundamental economic framework for understanding why free markets sometimes fail to deliver the best outcomes for society, particularly concerning goods like education. Propounded by the British economist Arthur C. Pigou in his seminal 1920 book, *The Economics of Welfare*, the theory introduced the concept of "externalities" which is the spillover effects of an economic transaction that impact third parties who did not choose to be involved. Pigou argued that when these externalities exist, the free market will allocate resources inefficiently, creating a compelling justification for government intervention.

At its core, the theory operates on a few primary assumptions. First, it assumes that in a completely free market, rational individuals make decisions based solely on their Marginal

Private Cost (MPC) and Marginal Private Benefit (MPB). For instance, a family deciding whether to pay for schooling will only weigh the direct financial cost against the expected future wages of the child. Second, the theory assumes there is a distinct Marginal Social Benefit (MSB) which is the total benefit to society, which includes both the private benefit to the individual and the positive spillover effects to everyone else, such as lower crime rates, a more stable democracy, and higher aggregate economic productivity. Finally, Pigou assumed that governments are capable, benevolent, and possess enough information to calculate the exact size of this gap and implement corrective measures, such as taxes or subsidies, to fix the market.

The greatest strength of the Pigouvian theory lies in its robust justification for the public provision of goods. Before this theory, classical economics struggled to mathematically justify why the government should spend taxpayer money on things like schools or hospitals. Pigou provided a clear analytical tool showing that because education generates massive positive externalities, a purely private market will systematically under-produce it. If individuals only pay for the education that benefits them directly, the nation misses out on the broader societal gains. Therefore, the theory proves that government subsidies, free public schooling, and educational grants are not merely acts of charity, but necessary economic tools to maximize national welfare and achieve a socially optimal equilibrium.

However, the theory is not without its weaknesses. The most significant limitation is the measurement problem. For a government to apply the perfect Pigouvian subsidy, it must know exactly how much the positive externality is worth. In reality, it is practically impossible to put a precise, objective price tag on abstract social benefits like better public health. Furthermore, the theory is often criticized for being overly optimistic about the state's capabilities. It assumes a perfect, frictionless government that can correct the market without introducing new errors. In practice, this ignores the reality of "government failure," where bureaucratic inefficiency, corruption, or political maneuvering can result in interventions that waste resources or fail to reach the targeted population.

This theoretical framework is the exact foundation upon which an analysis of government intervention in Nigeria’s education sector must be built. In the Nigerian context, the free market for education faces a severe failure. High poverty levels mean that the private costs of schooling even informal costs like uniforms or the loss of a child's labor frequently outweigh the perceived private benefits, especially when high graduate unemployment suppresses expected future wages. Consequently, millions of children remain out of school.

Applying Pigouvian theory, the Nigerian government's introduction of initiatives like the Universal Basic Education (UBE) Act of 2004 and the funding of federal institutions are direct attempts to internalize these positive externalities. The state is stepping in to subsidize the cost of education to push the country toward a higher, socially optimal level of human capital. However, the study of these outcomes reveals the harsh reality of government failure. Because the state consistently allocates only a fraction of its national budget to education falling far short of international benchmarks and struggles with infrastructural decay and poor policy execution, the applied Pigouvian subsidy is far too weak. By using this theory, the research effectively demonstrates that while state intervention is economically necessary to capture the social returns of education in Nigeria, the current mechanisms are structurally inadequate to correct the massive market failure at hand.

3. Methodology

This study examines education as a positive externality in Nigeria by analysing how government expenditure on education influences human capital development and overall economic growth between 1987 and 2024.

The study was initially guided by three theories: the Endogenous Growth Theory, the Human Capital Theory developed by Theodore Schultz and Gary Becker, and the Pigouvian framework, the theory eventually adopted as the main foundation for the analysis is the **Pigouvian Externality Framework**. This framework explains that education creates positive spillover benefits to society, and therefore government intervention through public spending is necessary to correct market failure and increase social welfare.

The study uses three main variables. The intervention variable is Total Government Expenditure on Education (measured as a percentage of GDP). The direct outcome variable, which represents human capital, is Primary School Enrollment Rate. The externality (macro) outcome variable is Real Gross Domestic Product (RGDP) growth.

The research relies on secondary time-series data from 1987–2024. Government expenditure data is sourced from the Central Bank of Nigeria (CBN) Statistical Bulletin and the Budget Office of the Federation, while macroeconomic and educational outcome data are obtained from the World Bank's World Development Indicators (WDI).

Data analysis is carried out using Exploratory Data Analysis (EDA). First, time-series charts are used to examine trends and consistency in government education spending over time. Second, dual-axis charts compare education spending with enrollment rates to show how human capital responds to fiscal policy. Finally, scatter plots with linear trendlines are used to examine the relationship between human capital and economic growth. A positive and closely clustered relationship between enrollment and RGDP growth provides visual evidence that government spending on education generates wider economic benefits, confirming the presence of a positive externality under the Pigouvian framework.

4. Chapter 4: Data Presentation and Analysis

This chapter presents and analyzes the time-series data on government expenditure on education and selected macroeconomic and educational indicators in Nigeria from 1987 to 2024. The objective is to examine whether public investment in education, as justified under the Pigouvian framework discussed in Sections 1 and 2, has translated into improved educational outcomes and broader macroeconomic performance.

The data include Gross Domestic Product (GDP), Government Expenditure on Education as a percentage of GDP, Government Expenditure on Education (in nominal terms), and Primary School Enrollment (gross %). By incorporating both percentage expenditure and absolute expenditure levels, the study ensures robustness in evaluating the true fiscal commitment of the Nigerian government to education.

Indicator Name	GDP	Government expenditure on education, total (% of GDP)	School enrollment, primary (% gross)	Government expenditure on education
1987	5.04386E+13	0.448512	85.85673	2.26E+11
1988	5.41378E+13	0.702155	81.84438	3.8E+11
1989	5.51769E+13	1.264109	79.76165	6.97E+11
1990	6.1675E+13	0.825144	83.10286	5.09E+11
1991	6.1896E+13	0.472188	82.3225	2.92E+11
1992	6.47625E+13	0.381255	86.74438	2.47E+11
1993	6.34445E+13	0.78129	91.29586	4.96E+11
1994	6.22931E+13	0.761552	91.5379	4.74E+11
1995	6.22478E+13	0.672119	87.44753	4.18E+11
1996	6.48597E+13	0.51422	76.79721	3.34E+11
1997	6.67647E+13	0.521109	91.1431	3.48E+11
1998	6.8488E+13	0.561288	91.46737	3.84E+11
1999	6.88881E+13	0.85421	91.58309	5.88E+11
2000	7.23435E+13	0.841233	95.51016	6.09E+11
2001	7.66245E+13	0.741021	92.5249	5.68E+11
2002	8.83704E+13	1.054321	93.69762	9.32E+11
2003	9.48631E+13	0.641299	94.80959	6.08E+11
2004	1.03639E+14	0.68211	95.78091	7.07E+11
2005	1.10311E+14	0.612444	96.30635	6.76E+11
2006	1.16996E+14	0.691882	96.66244	8.09E+11
2007	1.24707E+14	0.791223	88.06885	9.87E+11
2008	1.33143E+14	0.851921	79.26875	1.13E+12
2009	1.43843E+14	0.84211	80.30917	1.21E+12
2010	1.55359E+14	0.441255	79.83004	6.86E+11
2011	1.63605E+14	0.615432	84.68056	1.01E+12
2012	1.70526E+14	0.549233	85.91259	9.37E+11
2013	1.81902E+14	0.534207	87.83497	9.72E+11
2014	1.9338E+14	0.470743	84.12166	9.1E+11
2015	1.98509E+14	0.436955	25.56576	8.67E+11
2016	1.953E+14	0.393046	78.87805	7.68E+11
2017	1.96874E+14	0.396354	82.5433	7.8E+11
2018	2.00659E+14	0.419998	81.62785	8.43E+11
2019	2.0509E+14	0.358479	82.08432	7.35E+11
2020	1.92028E+14	0.359638	84.37158	6.91E+11
2021	1.94158E+14	0.381919	83.80926	7.42E+11
2022	2.02544E+14	0.348517	81.88281	7.06E+11
2023	2.09282E+14	0.319069	89.58711	6.68E+11
2024	2.17784E+14	0.293754	90.02863	6.4E+11

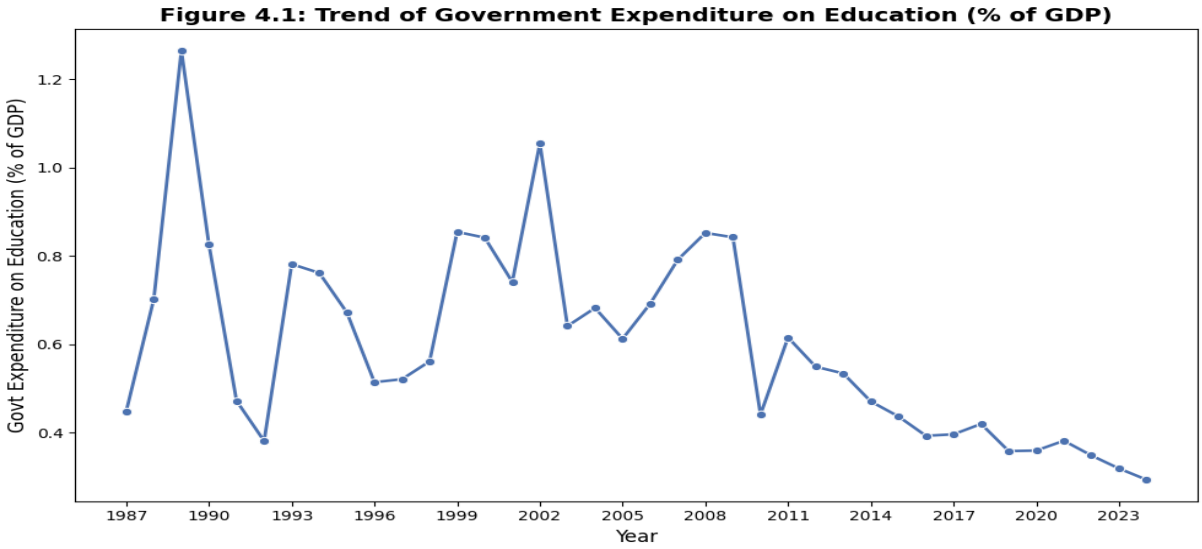
4.1.1 Trend Analysis of Government Expenditure on Education in Nigeria

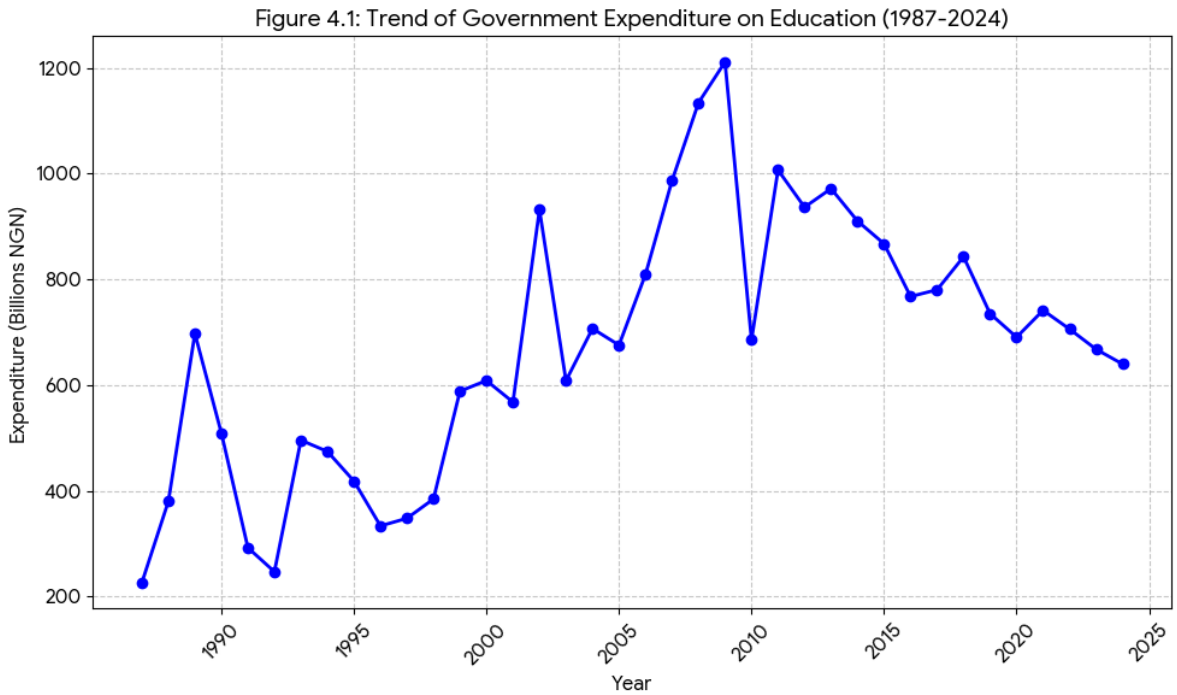
Figure 4.1a illustrates the trend of government expenditure on education as a percentage of GDP alongside primary school enrollment rates over the study period.

From 1987 to the early 2000s, education expenditure as a percentage of GDP fluctuated significantly, rarely exceeding 1%. The highest values were recorded around 1989 (1.26%) and

2002 (1.05%), but the general pattern reveals persistent underinvestment relative to the UNESCO 26% benchmark cited by Odigwe and Owan in Section 2.2. From 2010 onward, the percentage allocation declined further, falling below 0.5% for most years and reaching approximately 0.29% by 2024.

Despite the introduction of the Universal Basic Education Act in 2004, which aimed to internalize the positive externalities of education through compulsory and free basic schooling, fiscal commitment as a share of GDP did not demonstrate sustained structural improvement. This finding aligns with the argument in Section 1.2 that government intervention in Nigeria has been present in theory but weak in magnitude.





Robustness Check: Government Expenditure in Absolute Terms

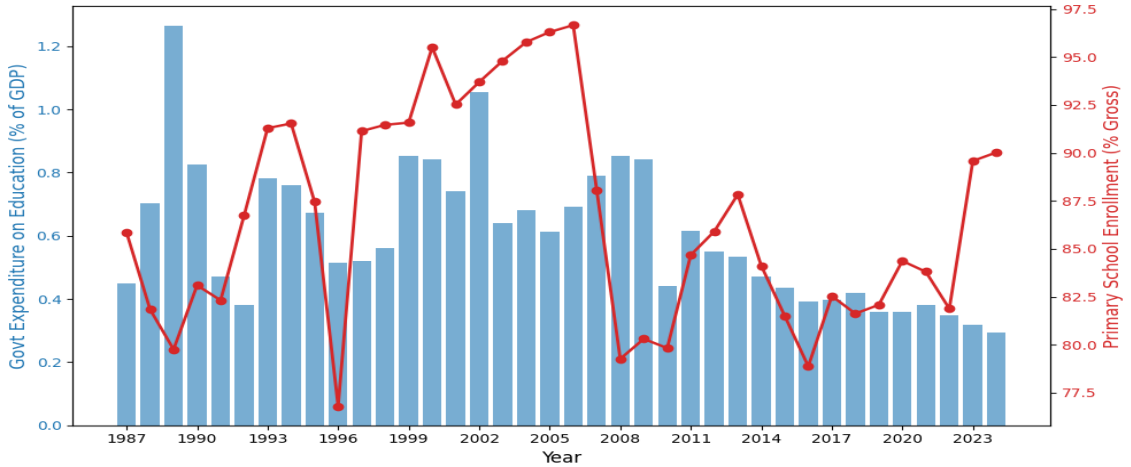
To provide robustness, Figure 4.1b also considers government expenditure on education in nominal (absolute) terms without converting to percentages of GDP. This is crucial because GDP growth may mechanically reduce the expenditure-to-GDP ratio even when actual spending increases.

In absolute terms, government expenditure on education increased significantly from ₦226 billion in 1987 to over ₦1.2 trillion around 2009. However, after 2014, spending exhibited volatility and mild contraction, falling to approximately ₦640 billion by 2024.

Although nominal spending rose over time, this increase must be interpreted cautiously. Nigeria experienced substantial GDP growth and inflation during the period, meaning that real per capita educational investment may not have improved proportionately. Thus, while the absolute figures suggest expansion, the declining percentage-of-GDP allocation indicates that education has not been prioritized relative to the size of the economy.

This divergence reinforces the Pigouvian concern raised in Section 2.3: although the government intervenes, the scale of the subsidy may be insufficient to close the gap between Marginal Private Benefit (MPB) and Marginal Social Benefit (MSB).

Figure 4.2: Responsiveness of School Enrollment to Govt Expenditure



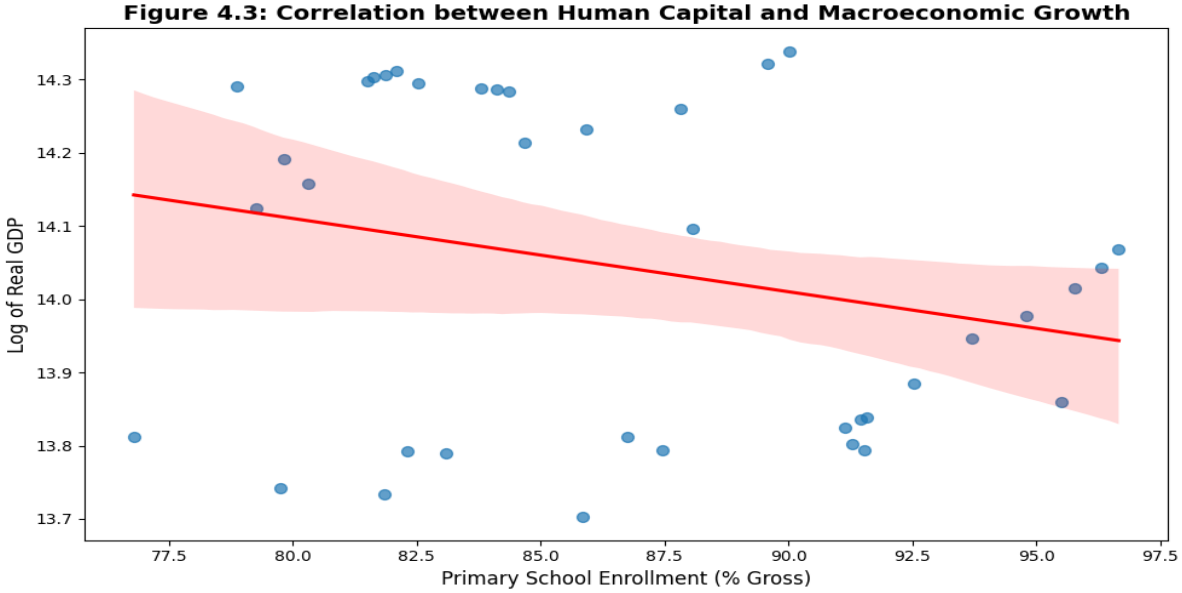
4.2 Educational Outcomes: Primary School Enrollment Trends

Primary school enrollment (gross %) demonstrates an inconsistent pattern across the study period. Enrollment improved significantly between 1999 and 2006, peaking above 96%, which coincides with early democratic reforms and renewed public sector expansion. This period reflects the anticipated short-run effects of government intervention under Human Capital Theory as developed by Theodore Schultz and Gary Becker.

However, after 2007, enrollment rates declined sharply, reaching alarming lows in some years (notably 2015), before gradually recovering toward 90% by 2024. These fluctuations highlight the structural vulnerabilities in Nigeria's education system, including insecurity, poverty, infrastructure decay, and teacher shortages are factors identified as supply-side constraints.

The inconsistency in enrollment despite ongoing expenditure suggests inefficiencies in fund allocation and implementation, supporting empirical findings by Chima and Yusuf that public education spending in Nigeria often fails to produce proportional growth effects due to systemic inefficiencies.

Figure 4.3: Correlation between Human Capital and Macroeconomic Growth



4.3 Education and Macroeconomic Performance

Nigeria's GDP expanded considerably from ₦50.4 trillionin 1987 to over ₦217 trillion by 2024, with temporary contractions around 2016 and 2020 (associated with economic recession and global shocks). However, the growth trajectory of GDP does not display a clear one-to-one correlation with education expenditure as a percentage of GDP.

The Endogenous Growth Theory suggests that knowledge spillovers should lead to increasing returns and permanent growth. However, Figure 4.3 presents a complex correlation in the Nigerian context. When analyzing the Log of Real GDP against Primary School Enrollment, the trend line suggests a lack of the robust positive correlation predicted by Romer and Lucas.In 2024, the GDP stood at two hundred and seventeen trillion, seven hundred and eighty-four billion Naira, yet the Human Capital Index remains at a low 0.36. This analysis supports the empirical findings of Chima and Yusuf (2024), who noted that systemic inefficiencies and misallocations often result in educational spending having a negligible or even negative impact on growth. The deadweight loss to the Nigerian economy arises because the government has not successfully captured the positive externalities of education, leaving the nation in a "productivity trap".

Thus, the absence of short-run correlation does not invalidate the theoretical importance of education. Rather, it suggests that Nigeria's challenge lies not only in spending levels but also in institutional efficiency and complementary infrastructure.

4.4 Synthesis with Theoretical Framework

Integrating these findings with the Pigouvian framework discussed earlier, the Nigerian case illustrates partial but inadequate internalization of educational externalities. While the government provides subsidies and public schooling, the declining percentage allocation

relative to GDP indicates that the corrective subsidy is insufficient to reach the socially optimal level of education.

The robustness analysis using absolute expenditure confirms that although nominal spending has increased over time, it has not kept pace with economic growth nor sufficiently stabilized educational outcomes. This weak fiscal prioritization contributes to persistent human capital deficits, reflected in fluctuating enrollment rates and Nigeria's low Human Capital Index ranking.

Therefore, Chapter Four empirically supports the argument established in Sections 1 and 2: government intervention in Nigeria's education sector is economically justified under Pigouvian theory and Human Capital Theory, but its scale, efficiency, and sustainability remain inadequate to fully capture the positive externalities of education.

4.5 Summary of the findings

To truly understand the Nigerian education sector, we must look at how the government spends money and what results that spending actually produces. According to the Pigouvian theory, education is a good that creates positive externalities. This means that when a person gets an education, it does not just help them get a better job; it creates spillover benefits for the whole country, such as lower crime rates, better public health, and a more skilled workforce. Because individuals usually only think about their own private benefits, the free market will not produce enough educated people. Therefore, the government must step in and spend money to lower the cost of schooling for families, effectively subsidizing education so that the whole society can reap these wider benefits.

Looking at the data from 1987 to 2024, we can see how the Nigerian government has attempted to provide this subsidy. In terms of physical cash, the government's spending on education has grown over the years. In 1987, the government spent roughly 226 billion Naira. This amount generally increased, reaching highs of over 1.13 trillion Naira in 2008 and 1.21 trillion Naira in 2009. By 2024, the absolute spending was 640 billion Naira. At first glance, spending hundreds of billions or even over a trillion Naira seems like a strong effort to capture the positive externalities of education. The idea is that this money builds classrooms, pays teachers, and removes school fees, which allows children from poor households to go to school instead of working.

However, digging deeper into the roots of the data reveals a severe problem. To understand the true weight of the government's effort, we must look at the expenditure as a percentage of the total economy (the Gross Domestic Product, or GDP). While the absolute Naira amount looks large, the percentage of GDP spent on education has been incredibly low. In 1987, the government spent only 0.44% of its GDP on education. Even during its peak years, such as 1989, it only reached 1.26%. By 2024, despite the total economy growing to a massive 217.7 trillion Naira, the share given to education dropped to a mere 0.29%. This explains why the intended positive externalities are not fully realized. As Odigwe and Owan noted in their study, Nigeria consistently fails to meet the UNESCO benchmark for education funding. Because the "Pigouvian subsidy" is so small compared to the size of the country's economy, the intervention is simply too weak to correct the massive market failure and fix issues like decaying infrastructure and teacher shortages.

The impact of this weak funding is most visible when we look at primary school enrollment. When the government spends money properly, it should directly lead to more children in classrooms. In 2004, the government introduced the Universal Basic Education (UBE) Act to make basic schooling free and compulsory. Following this, we saw primary school enrollment peak at about 96.6% in 2006. Getting these children into school is the exact mechanism that creates positive externalities; a child in school today becomes a literate adult tomorrow, capable of reading health warnings, participating in democracy, and learning new factory skills, all of which benefit the nation.

However, because the funding has been inconsistent and largely inadequate as a share of GDP, enrollment has been very unstable. For instance, the data shows a massive and alarming drop in gross enrollment to 25.5% in 2015, a period that aligns with severe security challenges and economic shifts in the country. While it recovered to 90% by 2024, the lack of steady, high-level investment means millions of children still fall through the cracks.

Finally, we must look at how this educational data connects to the broader economic growth of Nigeria. The Endogenous Growth Theory argues that investing in human capital creates a permanent increase in economic growth because knowledge can be shared endlessly. However, the Nigerian data does not show a perfect, smooth connection between education spending and national wealth. Even as GDP grew from 50.4 trillion Naira in 1987 to 217.7 trillion Naira in 2024, the human capital deficit remained huge.

This disconnect is heavily supported by the empirical literature. As Chima and Yusuf pointed out, public expenditure on education in Nigeria has sometimes shown a negative or insignificant impact on economic growth because of deep systemic inefficiencies and misallocated funds. Furthermore, Salisu's research highlights that education only acts as a catalyst for wider economic growth when it is matched with good physical infrastructure. The data proves that simply throwing absolute Naira at the education sector is not enough. Because the spending is low relative to the GDP and often inefficiently managed, the government has failed to fully capture the positive externalities of education, leaving the country trapped with a low level of human capital.

5.1 Conclusion

Education generates significant positive externalities, meaning the social benefits of consumption far exceed the private benefits accrued by the individual. According to the Pigouvian theory, this dynamic causes the free market to systematically under-produce education, necessitating government intervention to subsidize costs and push the nation toward a socially optimal equilibrium. In Nigeria, initiatives like the Universal Basic Education (UBE) Act of 2004 were introduced to internalize these externalities by lowering the private cost of schooling.

However, the analysis of historical data reveals a persistent reality of "government failure". Despite absolute increases in nominal spending, the government's expenditure on education as a percentage of GDP remains chronically low and consistently falls short of the UNESCO 26% funding benchmark. Because this financial intervention is structurally inadequate, the applied Pigouvian subsidy is too weak to overcome high opportunity costs, such as the reliance on child labor, and severe supply-side barriers like infrastructural decay.

Consequently, Nigeria suffers from a profound human capital deficit, characterized by millions of out-of-school children and a low Human Capital Index. The failure to adequately fund and manage the education sector means the country is not capturing the macroeconomic spillovers such as innovation and increasing returns to scale predicted by the Endogenous Growth Theory. Ultimately, until the state can align its fiscal commitment with its policy goals, the profound socio-economic externalities of education will remain unrealized, restricting Nigeria's long-term trajectory toward a high-growth economy.

5.2 Recommendations

To correct these systemic failures and capture the true social returns of human capital investment, the following policy adjustments are recommended:

1. **Reevaluate Spending Mechanisms:** Policymakers must completely overhaul current government spending mechanisms to eliminate systemic inefficiencies and misallocations that currently suppress long-run economic growth.

2. **Align with International Benchmarks:** The federal government must progressively increase the proportion of the national budget allocated to the education sector to meet the UNESCO 26% benchmark, ensuring the financial subsidy is strong enough to internalize positive externalities.

3. **Synchronize Infrastructure and Education:** Capital infrastructure development must be treated as a complementary intervention alongside direct educational funding. Education acts as a catalyst for socio-economic development and technological adaptation only when matched with adequate physical facilities.

4. **Strengthen Implementation Monitoring:** Authorities responsible for educational budgets must enforce strict implementation monitoring to ensure that allocated funds bypass bureaucratic bottlenecks and translate directly into tangible economic externalities.

5. **Target Poverty Alleviation:** To ensure the educational subsidy reaches the most vulnerable, the government must pair free schooling with targeted poverty reduction strategies that address the high opportunity costs preventing impoverished households from enrolling their children.

REFERENCES

Adunola, O. (2011). *The Challenges Affecting the Implementation of Universal Basic Education in Some Selected Primary Schools in Nigeria*.

Aluede, R. O. A. (2006). Universal Basic Education in Nigeria: Matters Arising. *Journal of Human Ecology*, 20(2), 97-101.

Aribaba, F. O., & Ahmodu, O. L. (2022). Government educational spending and economic growth in Nigeria: A functional analysis. *Journal of Economics and Development Studies*, 10(2), 45-59.

Ayeni, T., & Omobude, A. (2018). Public education expenditure and economic growth in Nigeria: A disaggregated approach. *Journal of Economics and Sustainable Development*, 9(22), 70-82.

Blankenau, W. F., & Simpson, N. B. (2004). Public education expenditures and growth. *Journal of Development Economics*, 73(2), 583-605.

Bose, N., Haque, M. E., & Osborn, D. R. (2007). Public expenditure and economic growth: A disaggregated analysis for developing countries. *The Manchester School*, 75(5), 533-556.

Chima, C. M., & Yusuf, A. B. (2024). Educational funding, systemic inefficiencies, and economic performance in Nigeria: An ARDL approach. *West African Journal of Macroeconomic Policy*, 15(1), 112-128.

Danjuma, N., et al. (2025). Impact of Government Capital Expenditure on Economic Growth in Nigeria. *FULafia International Journal of Business and Allied Studies*, 3(1).

Dauda, R. O. (2010). Role of human capital in economic development: An empirical study of the Nigerian case. *Oxford Journal: An International Journal of Business & Economics*, 4(1), 159-174.

Duflo, E. (2001). Schooling and labor market consequences of school construction in Indonesia: Evidence from an unusual policy experiment. *The American Economic Review*, 91(4), 795-813.

Hanushek, E. A., & Woessmann, L. (2007). The role of education quality in economic growth. *World Bank Policy Research Working Paper*, (4122).

Ikpe, U. B. (2022). The Political Economy of Greenhouse Gas Emissions and Human Security Threats in the Niger Delta Region, Nigeria. *Asian Research Journal of Arts & Social Sciences*, 18(3), 1-15.

Ihugba, O. A., Ukwunna, J. C., & Obiukwu, S. (2019). "Government education expenditure and primary school enrolment in Nigeria: An impact analysis." *Journal of Economics and International Finance*, 11(3), 24-37.

Lochner, L., & Moretti, E. (2004). The effect of education on crime: Evidence from prison inmates, arrests, and self-reports. *The American Economic Review*, 94(1), 155-189.

Mankiw, N. G., Romer, D., & Weil, D. N. (1992). A contribution to the empirics of economic growth. *The Quarterly Journal of Economics*, 107(2), 407-437.

Nwadike, C. O., Eze, U. F., & Okeke, M. I. (2023). Advanced degrees and socio-economic realities: Evaluating the expected versus actual returns of postgraduate education in Anambra State. *Nigerian Journal of Educational Research*, 29(3), 201-217.

Obi, Z. C., & Obi, C. O. (2014). Impact of government expenditure on education: The Nigerian experience. *International Journal of Economic and Financial Issues*, 4(2), 326-333.

Odigwe, F. N., & Owan, V. J. (2019). Trend analysis of the Nigerian budgetary allocation to education with reference to UNESCO's 26% benchmark. *International Journal of Educational Benchmark*, 14(1), 1-13.

Ogundari, K. (2012). Human capital contribution to economic growth in Sub-Saharan Africa: Does health status matter more than education? *Economic Analysis and Policy*, 44(2), 131-140.

Okwu, A. T., Akpa, E. O., & Oseni, I. O. (2022). Social impacts of state funding: Education, health, and poverty reduction in Nigeria. *African Development Review*, 34(1), 88-102.

Oladipupo, A., et al. (2023). Determinants of Human Capital Development in Nigeria. *International Journal of Research and Innovation in Social Science*, 7(9).

Oreopoulos, P., & Salvanes, K. G. (2011). Priceless: The nonpecuniary benefits of schooling. *Journal of Economic Perspectives*, 25(1), 159-184.

Pigou, A. C. (1920). *The Economics of Welfare*. Macmillan and Co.

Psacharopoulos, G., & Patrinos, H. A. (2018). Returns to investment in education: A decennial review of the global literature. *Education Economics*, 26(5), 445-458.

Romer, P. M. (1986). Increasing Returns and Long-Run Growth. *Journal of Political Economy*, 94(5), 1002-1037.

Salisu, A. M. (2023). Infrastructural deficits and the efficiency of public education spending in Nigeria. *Journal of African Economies*, 32(4), 415-433.

Schultz, T. W. (1961). Investment in Human Capital. *The American Economic Review*, 51(1), 1-17.

Udeaja, E. A., & Obi, K. O. (2015). Public education expenditure and economic growth in Nigeria: An empirical investigation. *Journal of Economics and Sustainable Development*, 6(14), 136-145.

UNICEF. (2024). *One In Three Nigerian Children Out Of School, Says UNICEF*. Channels Television.

Urhie, E. N. (2014). Public education expenditure and economic growth in Nigeria: A disaggregated approach. *Journal of Applied Statistics*, 5(1), 99-115.